Scheduling long-haul battery-electric trucks: coordinating charging, breaks, and daily rests

Steffen J.S. Bakker[a,*], Mostafa Mohammadi[a], Johan Stafseng[a], Magnus Stålhane[a], Odd André Hjelkrem[b]

[a]*Department of Industrial Economics and Technology Management, Norwegian University of Science and Technology, Alfred Getz veg 3, NO-7491 Trondheim, Norway*
[b]*Department of Energy Systems, SINTEF Energy Research - Trondheim, Norway*

Abstract

Recent advances in battery technology and fast charging are making long-haul battery-electric freight transport increasingly feasible. As vehicle ranges extend beyond regional distribution, charging decisions become increasingly intertwined with Hours-of-Service regulations. Since charging is time-consuming and unavoidable on long routes, operators must ensure that vehicles remain sufficiently charged while complying with driver regulations. Efficient operation therefore depends on synchronizing charging activities with mandatory driver breaks and daily rest periods. We study this scheduling problem for battery-electric trucks operating on multi-day long-haul routes. We introduce a multi-day electric truck driver scheduling problem in which charging decisions are optimized jointly with Hours-of-Service compliance to minimize total route duration. The problem is formulated as a resource-constrained shortest path problem and solved using an exact labeling algorithm that jointly models battery dynamics and driver regulations. Computational experiments show that optimized battery-electric truck schedules can, under favorable charging conditions, be completed within XX% of the duration of equivalent diesel-truck operations. However, schedule feasibility and performance deteriorate rapidly when charging opportunities are sparse or charging power is insufficient. Compared with a simpler charging strategy, the proposed optimization approach reduces route durations by up to XX%. Moreover, allowing partial recharging provides modest but consistent reductions in route time relative to full recharging. The results demonstrate that synchronization of charging and mandatory driver interruptions is a key operational enabler of long-haul freight electrification.

Keywords: Battery-electric trucks; long-haul freight transport; hours-of-Service regulations; charging scheduling; driver scheduling; resource-constrained shortest path.

1. Introduction

The electrification of road freight transport has progressed rapidly over the last decade. Battery-electric trucks are now being adopted in urban distribution and regional logistics, where vehicles can

*Corresponding author
Email address: steffen.bakker@ntnu.no (Steffen J.S. Bakker)

typically recharge at depots or terminals during predictable off-duty periods (Shoman et al., 2023). Recent improvements in battery capacity and fast-charging technology are extending feasible driving ranges and making battery-electric trucks increasingly viable for long-haul freight operations. Current commercial vehicle offerings illustrate this development: heavy-duty battery-electric trucks are commonly marketed with single-charge ranges of around 300 km, while newer long-haul models target approximately 500 km under favorable operating conditions (Volvo Trucks, 2025; Mercedes-Benz Trucks, 2025). However, road freight transport is a highly competitive business with low profit margins (Council of the European Union, 2025), where schedules are tight and time losses translate directly into cost. If charging causes additional time losses, battery-electric trucks can quickly become unattractive for long-haul freight operations.

A defining feature of long-haul trucking is that operations are constrained by Hours-of-Service (HoS) regulations. Under EU regulation, drivers must take mandatory breaks and daily rest periods, typically taken overnight, that reset driving and working-time limits (Goel & Irnich, 2017). Traditionally, compliance with these regulations has been the primary scheduling challenge, as diesel refueling is fast and widely available. Consequently, refueling is rarely modeled as a binding scheduling resource in conventional truck operations (Schulz, 2024). For battery-electric trucks, in contrast, charging is substantially more time-consuming than diesel refueling and depends on the availability of suitable charging infrastructure. Operators must therefore ensure that vehicles remain sufficiently charged while simultaneously complying with driver regulations. Efficient operation depends on synchronizing charging activities with mandatory breaks and daily rests such that charging can be absorbed into periods during which the driver is already required to be off duty.

In this paper, we investigate the synchronization of charging activities with mandatory breaks and daily rests in long-haul battery-electric freight transport. We study a multi-day scheduling problem in which charging, breaks, and daily rests must be jointly planned to minimize route duration while satisfying both Hours-of-Service and energy constraints.

1.1. Related literature

The Truck Driver Scheduling Problem (TDSP) concerns the construction of driver schedules that satisfy regulatory HoS constraints while optimizing operational objectives such as route duration or transportation cost. The Truck Driver Scheduling Problem (TDSP) has been studied extensively under Hours-of-Service regulations (Archetti & Savelsbergh, 2009; Goel, 2012; Goel & Kok, 2012).

Methodologically, TDSPs are commonly formulated as shortest path problems with resource constraints (SPPRCs). Feasible schedules are constructed incrementally through labeling algorithms that extend partial paths while tracking resources such as accumulated driving and working time since the last break or rest. Resource extension functions are used to update resource states and identify infeasible path extensions (Goel, 2010; Drexl & Prescott-Gagnon, 2010). Building on this paradigm, Goel & Irnich (2017) introduced an exact branch-and-price framework with a parameter-free network representation that avoids time discretization, while preserving regulatory feasibility. However, these models assume conventional vehicles with negligible refueling times and do not consider energy constraints.

While TDSP research focuses on regulatory feasibility, the Electric Vehicle Routing Problem (EVRP) literature provides a rich body of work on modeling battery dynamics and charging decisions (Schneider et al., 2014; Han et al., 2025). Many exact EVRP approaches are based on branch-and-price frameworks in which a master problem selects vehicle routes, while SPPRC pricing problems generate new feasible routes. In these formulations, battery state is treated as a resource and charging duration directly affect route timing (Desaulniers et al., 2016; Lam et al., 2022; Nafstad et al., 2025). Several works allow partial recharging and nonlinear charging functions to better reflect operational practice (Montoya et al., 2017; Froger et al., 2019). Whereas most EVRP studies focus on route construction and charging decisions along a route, recent work has begun to integrate routing decisions with the scheduling of charging infrastructure. For example, Yamín et al. (2025) jointly optimize urban delivery routes and overnight depot-charging schedules using a branch-price-and-cut algorithm. Their formulation accounts for nonlinear charging and limited charging capacity, but does not consider HoS-constrained driver schedules.

Only a limited number of studies explicitly consider the interaction between charging activities and driver breaks and rests under HoS regulation. Schiffer et al. (2017) extend the EVRP by incorporating HoS regulations and allowing en-route charging activities to be synchronized with mandatory driver breaks. Their EVRP is motivated by one-day routing and scheduling of electric commercial delivery vehicles in regional distribution networks. They solve the problem using a mixed-integer formulation combined with an adaptive large neighborhood search heuristic. Closer to the long-haul setting considered in this paper, Wan et al. (2024) study an opportunity-charging planning problem for a battery-electric truck in a single freight corridor case study. Their model coordinates charging decisions with planned stops at highway service areas and logistics hubs while accounting for charging technologies, electricity prices, and temporal flexibility. However, HoS regulations are represented only through simplified driving-time and rest constraints and the model does not consider recurring daily-rest requirements. More recently, Brockmann & Guajardo (2025) develop a mathematical programming model for an Electric Vehicle Truck Driver Scheduling Problem (EVTDSP) in which charging operations and mandatory driver breaks are jointly optimized. Their analysis highlights the importance of accounting for driver regulations when assessing the operational performance of battery-electric trucks. However, the study is restricted to single-day operations and does not consider recurring daily rests or multi-day scheduling decisions. Finally, Schneider et al. (2023) analyze one-day electric truck operations and evaluate predefined charging and EU HoS-complient break strategies, including alternatives enabled by high-power charging technologies such as the Megawatt Charging System. While their study provides valuable insights into the interaction between charging technology and driver regulations, it only evaluates a limited number of predefined operating strategies and does not address the underlying scheduling problem.

1.2. Research gap and contributions

It remains unclear to what extent charging can be absorbed within mandatory driver breaks and daily rests in multi-day long-haul operations, and how incomplete synchronization affects route duration and schedule feasibility. Existing studies focus primarily on distribution networks and single-day operations, whereas

multi-day operations involve repeated charging, mandatory breaks, and recurring daily rests, creating greater potential for both synergies and conflicts between energy and regulatory requirements. Furthermore, existing synchronization studies rely primarily on mixed-integer formulations and heuristic solution approaches and do not provide an exact method for jointly optimizing charging, breaks, and daily rests over multiple days.

To address this gap, we study a multi-day variant of the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which determines how charging activities, mandatory breaks, and daily rest periods should be scheduled and synchronized in long-haul battery-electric freight operations. Given a prescribed route and a set of charging opportunities, the objective is to minimize total route duration while satisfying both HoS and battery constraints.

The contributions of this paper are threefold. First, we extend the truck driver scheduling problem to the multi-day battery-electric setting by explicitly integrating charging decisions with both mandatory breaks and recurring daily rest periods. Second, we develop an exact resource-constrained shortest path formulation and labeling algorithm by extending the framework of Goel & Irnich (2017) with battery dynamics, tailored resource extension functions, and dominance rules. Third, we introduce an open-source benchmark set for the EVTDSP informed by interviews and operational data collected from long-distance trucking companies in Norway and Northern Europe, and use it to systematically analyze how route structure, charging conditions, and rest requirements affect schedule feasibility, charging-rest synchronization, and the value of partial recharging.

2. Problem definition

This paper studies the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which concerns the construction of HoS-compliant and energy-feasible schedules for a battery-electric truck operating on a long-haul route. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the timing of mandatory breaks and daily rests, as well as the timing, location, and amount of battery charging. Figure 3 illustrates the route structure considered in the problem: the truck travels from origin o to destination d through a predetermined sequence of Ncustomers, with charging stations available along the route. Let the predetermined route be represented by the ordered sequence of mandatory locations $\mathcal{R} = (c_0, c_1, \ldots, c_N, c_{N+1})$, where c_0 denotes the origin, c_{N+1} the destination, and the intermediate locations correspond to customer visits. Consecutive locations (c_i, c_{i+1}) define route legs. Moreover, let $\mathcal{L}$ denote the set of all route *legs*, i.e., $\mathcal{L} = \{(c_i, c_{i+1}) \mid i = 0, \ldots, N-1\}$. Charging stations may be located along a leg. Let $\mathcal{P}$ denote the ordered set of physical nodes obtained by inserting all charging stations into the route. Consequently, each leg may be decomposed into one or more *sublegs* connecting consecutive physical nodes in $\mathcal{P}$. introduce a short table with notation? There are some things in the paper not defined.

Service at each customer requires a fixed, non-interruptible duration. Distances and travel times between consecutive physical locations are assumed known. Energy consumption is proportional to distance traveled and independent of cargo load. The vehicle is equipped with a battery of fixed capacity and may recharge at

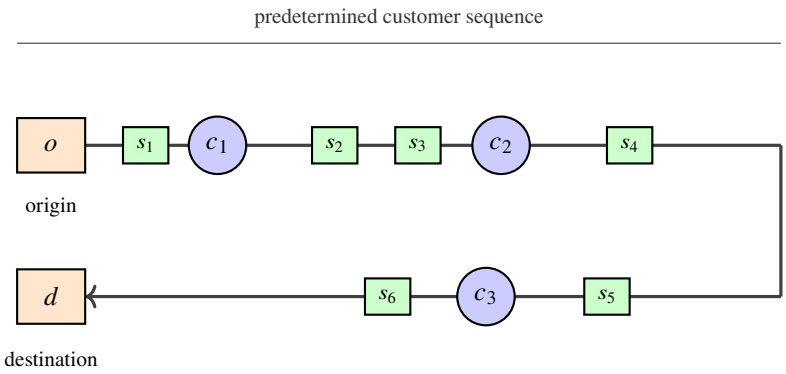


Figure 1: Illustrative route structure for the EVTDSP. The truck travels from origin o to destination d through a predetermined sequence of customers (c_1, c_2, c_3), with charging stations $(s_1, \ldots, s_6)$ available along the route.

charging stations located along the route. Charging stations are assumed homogeneous and provide energy at a fixed power level, implying a linear relationship between charging time and energy gained. At each charging station, the vehicle may recharge either to full capacity or by a selected amount of energy. Multiple charging stops are permitted throughout the route. Charging is not allowed during customer service.

The driver is subject to EU HoS regulations, which define allowable driving and working durations, as well as mandatory break and rest periods. Driving time includes only the operation of the vehicle between locations, while working time encompasses both driving and service time at customers. A break is a pause of at least 45 minutes required after a maximum of 4.5 hours of driving. A daily rest is an uninterrupted period of at least 11 hours required after a maximum of 9 hours of driving or 13 hours of total working time. Figure 4 illustrates a valid HoS-compliant schedule under these rules. While EU regulations provide additional flexibility through mechanisms such as split breaks , the base formulation focuses on standard 45-minute breaks and full 11-hour daily rests. Additional regulatory options can be incorporated within the proposed framework but are omitted to maintain focus on the core scheduling problem. REF to a goel paper doing more intricate HoS regulation?

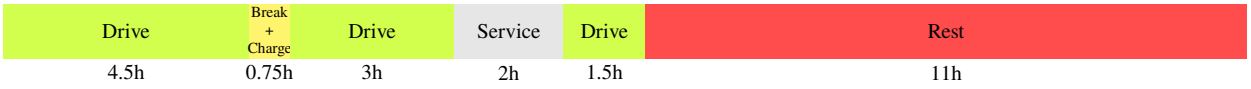


Figure 2: Example of a schedule of a fully rested driver in accordance with EU HoS regulation.

Breaks and daily rests cannot be scheduled during customer service but may be taken at customer locations, charging stations, or intermediate points along the route. Charging and service activities are mutually exclusive. Breaks and daily rests may be taken at charging stations, allowing charging activities to be synchronized with mandatory driver breaks and rests.

Along each route leg, the planner must decide whether to continue driving, initiate a mandatory break or daily rest. At each charging-station physical node, the planner additionally decides whether to charge and by how much. The objective is to minimize total route duration, consisting of driving time, customer service time, charging time, and mandatory breaks and daily rests.

3. Problem definition version 2

This paper studies the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which concerns the construction of HoS-compliant and energy-feasible schedules for a battery-electric truck operating on a long-haul route. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the timing of mandatory breaks and daily rests, as well as the when, where and how much to charge the battery. We consider a planning horizon of one week, denoted by $T_{\text{horizon}} = 168$ hours.

The problem can be modeled on a graph $G = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ and $\mathcal{A}$ is the set of nodes and arcs in the graph. The set of Nodes can be divided into one set of *customer nodes* $\mathcal{N}^C$ representing customer locations, and a set of charging nodes $\mathcal{N}^S$ representing the set of available charging stations along the route. In addition we have an origin node o to destination node d so that $\mathcal{N} = \mathcal{N}^C \bigcup \mathcal{N}^S \bigcup \{o, d\}$. As the sequence of the customer visits is pre-determined, and we only consider charging at stations along the route, the set of nodes have a unique order. Hence, without loss of generality we assume that the nodes are indexed according to this ordering, i.e., $\mathcal{N} = \{i_1, \dots i_{|\mathcal{N}|}\}$ where i_k represents the k-th node, and the set of arcs is simply $\mathcal{A} = \{(i_k, i_{k+1}) | k = 1, \dots, |\mathcal{N}| - 1\}$. Figure 3 illustrates one example of G for a route with three customer nodes and 6 charging stations.

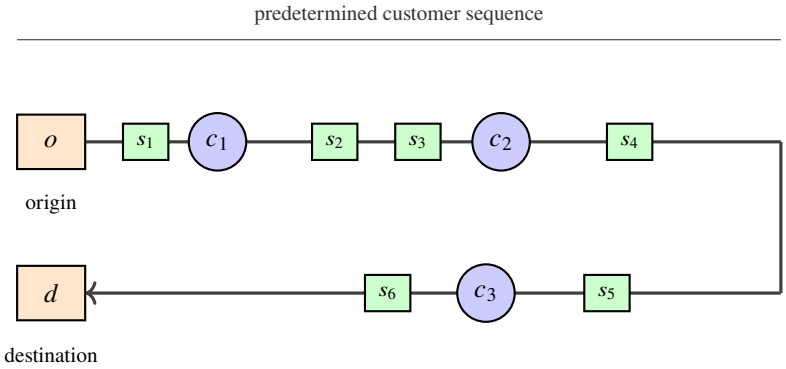


Figure 3: Illustrative route structure for the EVTDSP. The truck travels from origin o to destination d through a predetermined sequence of customers (c_1, c_2, c_3), with charging stations $(s_1, \dots, s_6)$ available along the route.

Each customer node $i \in \mathcal{N}^C$ has a fixed and uninterruptible service time s_i, while all charging nodes $i \in \mathcal{N}^S$ have a homogeneous charging rate ?? denoting the amount of energy added to the battery per time unit of charging. Furthermore, each arc $(i, j) \in \mathcal{A}$ has a parameter t_{ij} and ???, denoting the amount of time and energy used to traverse the arc, respectively. The vehicle is equipped with a battery of fixed capacity, denoted B, and may recharge, either partially or fully, at any number of charging stations located along the route.

The driver is subject to EU HoS regulations, which define allowable driving and working durations, as well as mandatory break and rest periods. Driving time includes only the operation of the vehicle between locations, while working time encompasses both driving and service time at customers. A break is a pause of at least t_{break} hours required after a maximum of $t_{\texttt{drive}|B}$ hours of driving or $t_{\texttt{elapsed}|B}$ hours of total working time. A daily rest is an uninterrupted period of at least t_{rest} hours required after a maximum of

$t_{\texttt{drive}|R}$ hours of driving or $t_{\texttt{elapsed}|R}$ hours of total working time. Breaks and daily rests cannot be scheduled during customer service but may be taken at customer locations, charging stations, or intermediate points along the route. If a break or a daily rest is taken at a charging station, it may be synchronized with charging the vehicle. The objective is to determine the minimal possible route duration, while adhering to the HoS regulations, and ensuring that there is sufficient power in the vehicle's battery.

Figure 4 illustrates a valid HoS-compliant schedule under these rules. While EU regulations provide additional flexibility through mechanisms such as split breaks , the base formulation focuses on standard 45-minute breaks and full 11-hour daily rests. Additional regulatory options can be incorporated within the proposed framework but are omitted to maintain focus on the core scheduling problem. REF to a goel paper doing more intricate HoS regulation?

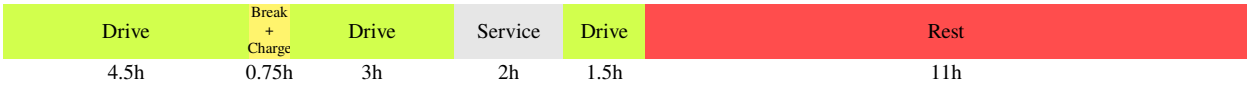


Figure 4: Example of a schedule of a fully rested driver in accordance with EU HoS regulation.

4. SPPRC Framework

The EVTDSP can be formulated as a shortest path problem with resource constraints (SPPRC) and solved using a labeling algorithm. Our modeling approach extends the parameter-free network and labeling framework of Goel & Irnich (2017) to account for battery-electric truck operations. Feasibility is tracked through label resources representing elapsed time, Hours-of-Service counters, and battery-energy state. These resources are propagated through the network by resource extension functions (REFs) associated with the network arcs. The framework supports both full- and partial-recharge strategies. We focus on the more general partial-recharge formulation, in which charging amounts are decision variables, while the full-recharge formulation is obtained as a special case and reported in Appendix A.

The remainder of this section first presents the underlying activity network, then introduces the label resources and REFs, and finally describes the feasibility and dominance rules used within the labeling algorithm.

4.1. Parameter-free network structure

Charging stations are introduced as additional physical nodes along the route, so that a leg between two mandatory locations may be decomposed into several consecutive sublegs. Figure 5 illustrates a route segment from customer n to customer m with a single intermediate charging station s, resulting in the sublegs (n, s) and (s, m).

Each subleg between two consecutive physical nodes n and m is represented by an activity subnetwork containing two intermediate states, denoted by $\tilde{n}^{\text{fit}}_{nm}$ and $\tilde{n}^{\text{dull}}_{nm}$. The *fit* state represents a situation in which the driver is allowed to continue driving, either because no break or daily rest is currently due or because the required one has just been completed. The *dull* state represents a situation in which the driver must first

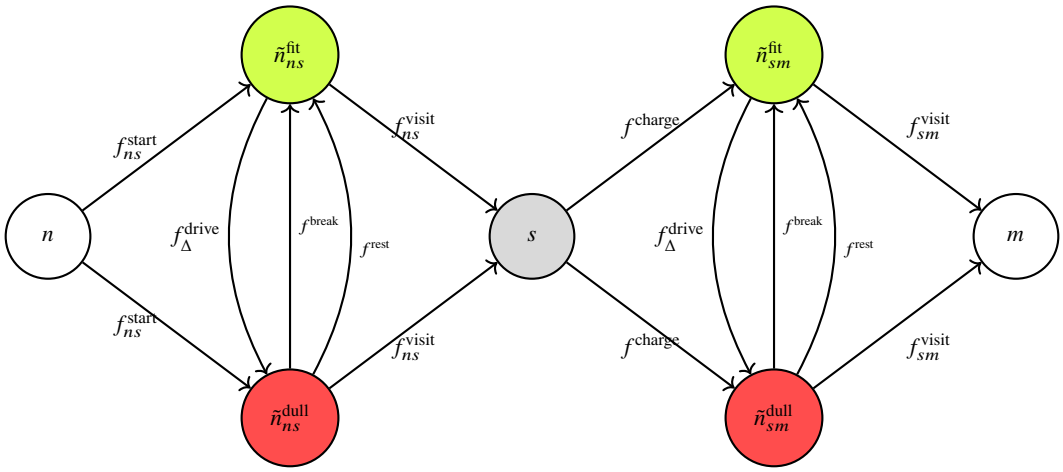


Figure 5: Extended subnetwork with an intermediate charging station s under the partial-recharge strategy. Arcs define resource extension functions.

take a break or daily rest before additional driving is permitted. Arcs between these states correspond to elementary driver activities, namely starting a leg, driving, taking a break, taking a daily rest, visiting the next physical node, and recharging at a charging station.

At a charging-station node, the schedule may initiate a charging activity. Charging is represented through recharge arcs originating from the charging-station node, allowing the schedule to continue after recharging with different battery-energy levels.

The resulting network contains six classes of arcs: start arcs, driving arcs, break arcs, rest arcs, visit arcs, and recharge arcs. A key feature of this representation is that a driving arc is not assigned a fixed duration in advance. Instead, its duration Δ is computed as the longest feasible driving duration given the current resource state. Consequently, each driving extension terminates either at the next physical node or at the point where HoS or energy constraints require a new decision.

4.2. Labeling algorithm

We solve the SPPRC using a labeling algorithm. The algorithm starts with an initial label at the origin, representing a fully rested driver and a fully charged vehicle. At each iteration, an unprocessed label is extended along every outgoing arc that is feasible given its current resource state. The corresponding resource extension function (REF) updates the temporal, regulatory, and energy resources. Labels violating an HoS or battery constraint are discarded, while dominance rules eliminate labels that cannot lead to a better feasible completion than another label at the same node. A complete schedule is obtained when a label reaches the destination.

4.2.1. Label definition

A label represents a partial path from the origin to the current network node, together with the resource state needed to assess its feasibility and extend it into a complete schedule. Formally, a label is represented

by

$$\mathcal{L} = (v, \pi, \mathbf{T}),$$

where v is the current network node, π stores the predecessor information required to reconstruct the partial path, and $\mathbf{T}$ contains its associated resources.

The resource vector includes the elapsed schedule time (`time`); the remaining driving time to the next physical node (`timeToNext`); the driving time since the last daily rest (`drive|R`); the driving time since the last break or daily rest (`drive|B`); the elapsed time since the end of the last daily rest (`elapsed|R`); the elapsed time since the end of the last break or daily rest (`elapsed|B`); and the vehicle's state of charge (`SOC`). The first six resources describe the temporal and HoS state of the partial schedule, while `SOC` describes its energy state. Table 1 summarizes the resources and their feasibility windows.

Table 1: Label resources with corresponding feasibility windows.

Resource	**Type**	**Description**	**Feasibility Window**
`time`	Temporal	Total time elapsed since departure	$[0, T_{\text{horizon}}]$
`timeToNext`	Temporal	Remaining driving time to next physical node	$[0, t_{nm}]$
`drive|R`	Regulatory	Driving time since last daily rest	$[0, t_{\texttt{drive}\mid R}]$
`drive|B`	Regulatory	Driving time since last break or rest	$[0, t_{\texttt{drive}\mid B}]$
`elapsed|R`	Regulatory	Elapsed time since last daily rest	$[0, t_{\texttt{elapsed}\mid R}]$
`elapsed|B`	Regulatory	Elapsed time since last break or rest	$[0, t_{\texttt{elapsed}\mid B}]$
`SOC`	Energy	Remaining battery energy (state of charge)	$[0,\ B]$

Under partial recharge, the amount of energy charged is a decision variable. Because charging time is linear in the amount of energy added, the feasible temporal and energy states can be represented by their interval endpoints, following interval-based representations in the literature (e.g., Desaulniers et al. (2016)). Accordingly, `time` and `SOC` are represented by $[l^{\min}_{\texttt{time}}, l^{\max}_{\texttt{time}}]$ and $[l^{\min}_{\texttt{SOC}}, l^{\max}_{\texttt{SOC}}]$, respectively, while the HoS-related resources remain scalar-valued. The resulting resource vector is

$$\mathbf{T} = \left(l^{\min}_{\texttt{time}}, l^{\max}_{\texttt{time}}, l_{\texttt{timeToNext}}, l_{\texttt{drive}\mid\texttt{R}}, l_{\texttt{drive}\mid\texttt{B}}, l_{\texttt{elapsed}\mid\texttt{R}}, l_{\texttt{elapsed}\mid\texttt{B}}, l^{\min}_{\texttt{SOC}}, l^{\max}_{\texttt{SOC}}\right).$$

4.2.2. Resource extension functions

Each arc $a = (i, j) \in A$ is associated with a REF f_a, which computes the resource vector $\mathbf{T}' = f_a(\mathbf{T})$, or valid lower bounds on its components. Extending a label $\mathcal{L} = (i, \pi, \mathbf{T})$ along arc a then produces the label $\mathcal{L}' = (j, \pi', \mathbf{T}')$, where π' contains the updated predecessor information. Furthermore, l'_r denotes the value of resource r after an extension, while l_r denotes its value before the extension.

The activity network uses six REFs: f^{start}_{ij}, $f^{\text{drive}}_{\Delta}$, f^{break}, f^{rest}, f^{visit}_{ij}, and f^{charge}_{i}. The first five build on the parameter-free framework of Goel & Irnich (2017). The temporal and HoS REFs are retained, while the driving REF is augmented to account for battery consumption. The recharge REF is introduced to model charging at charging stations.

The start REF initializes a subleg by setting the remaining travel time to t_{ij}. The drive REF advances

the schedule by a duration Δ, updating time, HoS counters, and battery energy. The break and rest REFs add the corresponding interruption and reset the applicable HoS counters. The visit REF records arrival at physical node j and, if j is a customer, adds the service time s_j.

The duration of a driving extension is determined endogenously as

$$\Delta = \min\Big\{l_{\text{timeToNext}}, t_R^{\text{drive}} - l_{\text{drive|R}}, t_B^{\text{drive}} - l_{\text{drive|B}}, t^{\text{day}} - \big(l_{\text{elapsed|R}} + t_{\text{rest}}\big)\Big\},$$

Thus, a driving extension ends when the next physical node is reached or when a break, daily rest, or daily working-time limit becomes binding. The drive REF reduces the SOC by $\alpha\Delta$, and the extension is feasible only if the resulting SOC remains nonnegative.

Under partial recharge, f_i^{charge} propagates the interval-valued time and SOC resources. Charging may add any amount of energy up to a full recharge, with charging time proportional to the energy added at rate CP. If charging overlaps with an admissible break or daily rest, only the unsynchronized charging time increases the schedule duration. Table 2 summarizes the resulting resource updates.

Table 2: Resource updates for the REFs on subleg $a = (i, j)$. A dash indicates that the corresponding resource remains unchanged. In the charge column, intervals give the updated lower and upper endpoints, respectively.

Updated resource	**Resource extension function**					
	f_{ij}^{start}	$f_{\Delta}^{\text{drive}}$	f^{break}	f^{rest}	f_{ij}^{visit}	f_i^{charge}
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_j$	$\begin{cases} \big[l_{\text{time}}^{\min}, l_{\text{time}}^{\max}\big] & \text{if } l_{\text{elapsed\|R}} = 0, \\ \big[l_{\text{time}}^{\min} + \max\{0, \tau^{\text{full}} - t_{\text{break}}\}, l_{\text{time}}^{\max}\big] & \text{if } l_{\text{elapsed\|B}} = 0, \\ \big[l_{\text{time}}^{\min}, l_{\text{time}}^{\max} + \tau^{\text{full}}\big] & \text{otherwise.} \end{cases}$
$l'_{\text{timeToNext}}$	t_{ij}	$l_{\text{timeToNext}} - \Delta$	—	—	—	t_{ij}
$l'_{\text{drive\|R}}$	—	$l_{\text{drive\|R}} + \Delta$	—	0	—	—
$l'_{\text{drive\|B}}$	—	$l_{\text{drive\|B}} + \Delta$	0	0	—	—
$l'_{\text{elapsed\|R}}$	—	$l_{\text{elapsed\|R}} + \Delta$	$l_{\text{elapsed\|R}} + t_{\text{break}}$	0	$l_{\text{elapsed\|R}} + s_j$	$\begin{cases} 0 & \text{if } l_{\text{elapsed\|R}} = 0, \\ l_{\text{elapsed\|R}} + \tau_{\text{add}}^{\min} & \text{otherwise.} \end{cases}$
$l'_{\text{elapsed\|B}}$	—	$l_{\text{elapsed\|B}} + \Delta$	0	0	$l_{\text{elapsed\|B}} + s_j$	$\begin{cases} 0 & \text{if } l_{\text{elapsed\|B}} = 0, \\ l_{\text{elapsed\|B}} + \tau_{\text{add}}^{\min} & \text{otherwise.} \end{cases}$
l'_{SOC}	—	$l_{\text{SOC}} - \alpha\Delta$	—	—	—	$\begin{cases} [B, B] & \text{if } l_{\text{elapsed\|R}} = 0, \\ [B, B] & \text{if } l_{\text{elapsed\|B}} = 0 \text{ and } \tau^{\text{full}} \le t_{\text{break}}, \\ \big[\min\{B, l_{\text{SOC}}^{\min} + CP\, t_{\text{break}}\}, B\big] & \text{if } l_{\text{elapsed\|B}} = 0 \text{ and } \tau^{\text{full}} > t_{\text{break}}, \\ \big[l_{\text{SOC}}^{\min}, B\big] & \text{otherwise.} \end{cases}$

For arc $a = (i, j)$, t_{ij} denotes the driving time from node i to j, and s_j denotes the service time at customer j. Parameters t_{break} and t_{rest} denote the mandatory break and daily-rest durations, respectively; α is the energy consumption rate; B is battery capacity; and CP is charging power. For the common REFs, each expression in the rows for l_{time} and l_{SOC} applies to both interval endpoints, unless the recharge column states otherwise. In the recharge column, comma-separated expressions denote the updates of $l^{\min}$ and $l^{\max}$,

respectively. Let $e^{\max} := B - l^{\min}_{\text{SOC}}$ and $\tau^{\text{full}} := e^{\max}/CP$. The added charging times in f_i^{charge} are

$$\tau^{\min}_{\text{add}} = \begin{cases} 0 & l_{\text{elapsed|R}} = 0, \\ 0 & l_{\text{elapsed|B}} = 0,\ \tau^{\text{full}} \le t_{\text{break}}, \\ \max\{0,\ \tau^{\text{full}} - t_{\text{break}}\} & l_{\text{elapsed|B}} = 0,\ \tau^{\text{full}} > t_{\text{break}}, \\ 0 & \text{otherwise}, \end{cases} \qquad \tau^{\max}_{\text{add}} = \begin{cases} 0 & l_{\text{elapsed|R}} = 0, \\ 0 & l_{\text{elapsed|B}} = 0,\ \tau^{\text{full}} \le t_{\text{break}}, \\ \tau^{\text{full}} - t_{\text{break}} & l_{\text{elapsed|B}} = 0,\ \tau^{\text{full}} > t_{\text{break}}, \\ \tau^{\text{full}} & \text{otherwise}. \end{cases}$$

The scalar full-recharge REFs in Table A.8 are recovered when $l^{\min} = l^{\max}$ and each recharge extension performs a complete recharge, so that $\tau^{\text{full}} = (B - l_{\text{SOC}})/CP$ and both interval endpoints coincide.

4.2.3. Feasibility and dominance

After each resource extension, the resulting label must remain feasible. Feasibility is enforced through the resource windows in Table 1; labels violating any bound are discarded immediately. Under partial recharge, the windows for `time` and `SOC` apply to each interval endpoint. In addition, the activity network restricts which extensions are allowed at each state: from a *dull* state, only break or rest arcs may be taken before driving resumes, whereas from a *fit* state driving is allowed and, once $l_{\text{timeToNext}} = 0$, visit or recharge arcs become available.

To limit the number of labels stored at each network node, dominance is applied among labels ending at the same node v. Let $\mathcal{L}_1 = (v, \pi_1, \mathbf{T}_1)$ and $\mathcal{L}_2 = (v, \pi_2, \mathbf{T}_2)$ be two feasible labels at node v. Label $\mathcal{L}_1$ dominates $\mathcal{L}_2$ if and only if:

$$\begin{aligned} l^{(1)}_{\text{timeToNext}} &\le l^{(2)}_{\text{timeToNext}}, & & \\ l^{(1)}_{\text{drive|R}} &\le l^{(2)}_{\text{drive|R}}, & l^{(1)}_{\text{drive|B}} &\le l^{(2)}_{\text{drive|B}}, \\ l^{(1)}_{\text{elapsed|R}} &\le l^{(2)}_{\text{elapsed|R}}, & l^{(1)}_{\text{elapsed|B}} &\le l^{(2)}_{\text{elapsed|B}}, \\ l^{(1)}_{\text{latest|R}} &\ge l^{(2)}_{\text{latest|R}}, & l^{(1)}_{\text{latest|B}} &\ge l^{(2)}_{\text{latest|B}}, \\ l^{\min,(1)}_{\text{time}} &\le l^{\min,(2)}_{\text{time}}, & & \\ l^{\min,(1)}_{\text{SOC}} &\ge l^{\min,(2)}_{\text{SOC}}, & l^{\max,(1)}_{\text{SOC}} &\ge l^{\max,(2)}_{\text{SOC}}. \end{aligned}$$

and, at the earliest departure time of label 2, label 1 must offer at least the same battery energy, accounting for optional charging between $l^{\min,(1)}_{\text{time}}$ and $l^{\max,(1)}_{\text{time}}$. Since the objective is to minimize total route duration, a label is preferable if its resource values are at least as good for continuing the route. For the scalar HoS resources (`timeToNext`, `drive|R`, `drive|B`, `elapsed|R`, and `elapsed|B`), which are monotone non-decreasing along every REF except break, rest, and synchronized recharge, label 1 dominates label 2 at the same node if each of these resources is no larger in label 1 than in label 2. This pairwise comparison is sufficient for SPPRC with monotone resources and preserves optimality (Irnich & Desaulniers, 2005; Goel & Irnich, 2017). Resources `latest|R` and `latest|B` are compared in the reverse direction because larger

values correspond to greater remaining scheduling flexibility. The partial-recharge extension introduces interval-valued time and SOC resources. The additional dominance conditions on $l^{\min}_{\text{time}}$, $l^{\min}_{\text{SOC}}$, $l^{\max}_{\text{SOC}}$, and the battery level at $t = l^{\min,(2)}_{\text{time}}$ follow the interval dominance rule for delivery paths with optional recharging (Desaulniers et al., 2016).

5. Experimental design

This section outlines the experimental design used in the computational study. We first describe the benchmark instances used to evaluate the proposed framework. We then introduce a baseline scheduling heuristic, which serves as a reference for assessing the benefits of jointly optimizing charging, breaks, and daily rests.

5.1. Benchmark instances

Long-haul freight operations differ substantially in their characteristics and operating conditions. Because the value of synchronizing charging with mandatory breaks and daily rests depends strongly on these factors, computational studies must be conducted under diverse operating conditions. We therefore develop benchmark instances that vary in route length, density of customers, and the distances between consecutive route legs, and combine these with scenario parameters describing charging-station spacing, battery capacity, charging power, and energy consumption. Rather than reproducing specific observed routes, the benchmark instances are designed to represent a broad range of realistic long-haul freight operations and constitute, to the best of our knowledge, the first benchmark set for the long-haul EVTDSP.

To ensure that the benchmark design reflects realistic long-haul freight operations, we conducted interviews with a diverse set of freight transport operators operating in Scandinavia and Northern Europe. The companies were identified with the assistance of the Norwegian Trucking Association (NLF), which helped ensure representation across geographical regions and transport segments. In total, 28 companies were contacted and interviews were completed with eight of them. The companies operate in several freight segments, including general freight-, container-, seafood-, temperature-controlled-, waste-, and bulk-transport.

The interviews revealed substantial variation in route characteristics. A typical route ranged from approximately 1000 to 1700 km, often requiring two to three days of operation and one or more overnight rests. Some routes consisted of direct movements between a small number of terminals or customers, whereas others involved multiple pickups and deliveries distributed across a region. Customers were frequently geographically clustered, producing route structures that combine long intercity transport legs with local collection and distribution activities. The interviews also highlighted the importance of charging-station availability and the potential for synchronizing charging activities with mandatory driver breaks and daily rest periods. These insights informed the benchmark instance design described in this subsection.

Based on these industry inputs, we develop an instance generator for heterogeneous long-haul routes. The generator first defines route instances along three dimensions: route length, customer distribution, and the number of customer stops. These dimensions are represented through route, cluster, and customer

classes, respectively. Scenario parameters are added subsequently to vary charging-station spacing, battery capacity, charging power, and energy consumption without changing the underlying route and customer structure.

Routes are classified as short, medium, or long, corresponding to increasing route distances. Specifically, short routes span 800–1200 km and typically require 1–2 overnight rests, while medium routes span 1500–2500 km and require 3–4 overnight rests. Long routes span 3000–4000 km and require 5–6 overnight rests, reaching the maximum multi-day operating horizon of six days before a longer weekly rest becomes necessary under the HoS regulations. For each instance, the total route length D_{total} is sampled uniformly from the interval associated with its route class.

Each route contains between one and three industrial clusters, interpreted as groups of geographically proximate customers. Cluster centers are positioned along intermediate sections of the route, so that customer visits are concentrated around these clusters and the first and last long-haul legs are typically the longest.

Customer stops are generated in two steps. First, one of three customer-count categories is selected: few (1–3 customers), medium (4–5 customers), or many (6–15 customers), and the total number of customers N is sampled from the selected range. Second, the customers are assigned to the route clusters, with at least one customer in each cluster. Customer locations are then placed near their cluster center, within 75 km in either direction along the route, while keeping local customer-to-customer distances between 30 and 150 km. This yields routes with long intercity legs between clusters and shorter local movements within clusters. Service times are varied with the customer-count categories to reflect different stop patterns. Instances with few customers use service times of 20–30 minutes per stop, medium instances use 10–20 minutes, and many-customer instances use 5–10 minutes.

For each combination of route class, number of clusters, and customer-count category, we generate two variants by resampling the route length and customer allocation. Instance names follow the format `routeClass_cl#clusters_cu#customers_v#variant`, for example `long_cl2_cu5_v1`.

After generating the structural route instances, we combine each route with scenario parameters. These parameters vary the availability of chargers along the route through the charging-station spacing, as well as battery capacity, charging power, and energy consumption. The parameter ranges are chosen to span both currently available heavy-duty battery-electric truck technologies and plausible near-future developments in vehicle and charging capabilities. Battery capacity ranges from 480 to 624 kWh, with an average of 540 kWh (Winkler et al., 2026). Energy consumption is set to 1.15 kWh/km, with a variation of ±0.15 kWh/km to account for different driving conditions and loads (Le Corguillé et al., 2025). REFS to reports for battery capacity, charging power, and energy-consumption ranges, if possible @mostafa/@Odd-Andre. The average driving speed is fixed to XXkmh -> Mostafa. across all instances based on historical data provided by one of the industry partners.

Charging stations are inserted along each route leg based on the selected charging-station spacing parameter. The range of spacing values is based on regulatory targets for publicly available charging infrastructure for heavy-duty vehicles under Regulation (EU) 2023/1804 (AFIR), which states that by December 2030, recharging pools must be deployed in each direction of travel with a maximum distance

of 60 km between them along the TEN-T core road network. For the TEN-T comprehensive road network, the maximum distance is 100 km (European Parliament and Council of the European Union, 2023). REF to some EU report stating min required distance between charging stations. . If a route leg is shorter than the specified spacing, a single charging station is placed at its midpoint. Otherwise, multiple charging stations are distributed along the leg such that the distance between consecutive charging opportunities does not exceed the specified spacing. Consequently, the charging-station spacing should be interpreted as a maximum spacing level rather than an exact distance between neighboring stations. Table 3 summarizes the instance classes, scenario values, and fixed HoS parameters used throughout the experiments.

Other parameters which are forgotten / not defined?

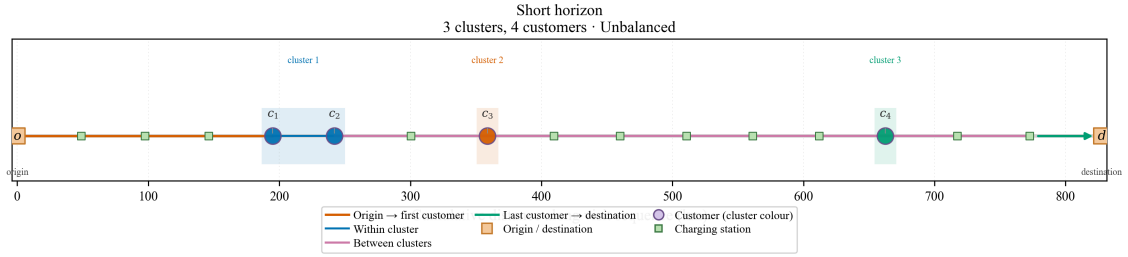
Table 3: Benchmark design: structural instance classes, scenario parameters, and fixed HoS constants.

Category	**Component**	**Values**
Structural instance classes		
Route class	Short	800–1200 km (1–2 nights)
	Medium	1500–2500 km (3–4 nights)
	Long	3000–4000 km (5–6 nights)
Clusters per route	Cluster class	1–3 clusters
Customer class	Few	1–3 customers
	Medium	4–5 customers
	Many	6–15 customers
Sampled customers	By route class	Short: 2–12; Medium: 2–13; Long: 2–14
Service time per stop	Few	20–30 min
	Medium	10–20 min
	Many	5–10 min
Variants	Per structural class	2 resampled variants
Scenario parameters, values for the base case marked bold		
Battery capacity	B	350, **500***, 750 kWh
Charging power	P_{ch}	**350***, 1000 kW
Charging-station spacing	d_{CS}	40, **50***, . . . , 100 km
Energy consumption	α	1.12, **1.49***, 2.08 kWh/km
Fixed HoS parameters		
Maximum continuous driving time	$t_{\text{drive}\mid B}$	4.5 h
Daily driving limit	$t_{\text{drive}\mid R}$	9 h
Maximum elapsed time before break	$t_{\text{elapsed}\mid B}$	6 h
Maximum elapsed time before daily rest	$t_{\text{elapsed}\mid R}$	13 h
Minimum break duration	t_{break}	0.75 h
Minimum daily rest duration	t_{rest}	11 h

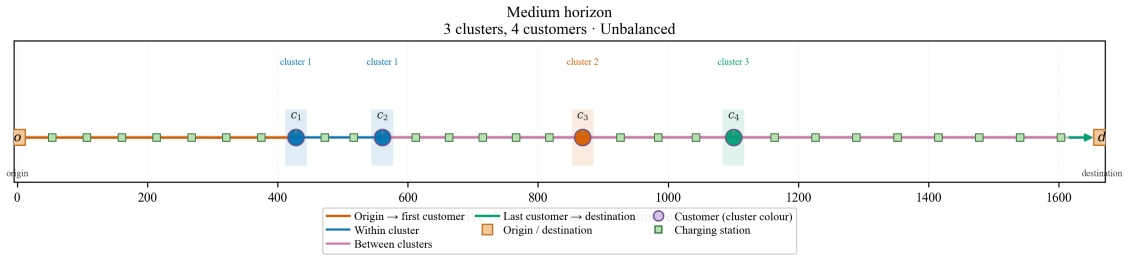
do we have a pseudocode summarizing the procedure? Add in appendix?

A total of 42 unique structural benchmark instances are generated by combining the route, cluster, and customer classes described above. Figure 6 graphically illustrates three representative instances under the

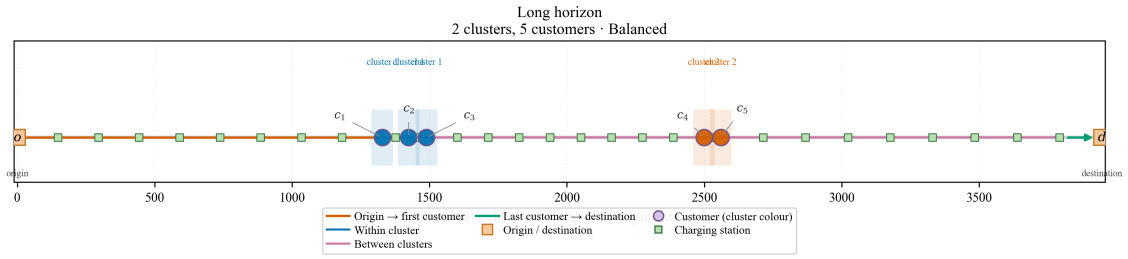
base scenario setting, highlighting the key dimensions of variation in route length and customer distribution. The full set of benchmark instances and their characteristics is reported in B.9.



(a) Short horizon, unbalanced.



(b) Medium horizon, unbalanced.



(c) Long horizon, balanced.

Figure 6: Three representative benchmark instances illustrating variation in route length and customer distribution.

5.2. Baseline heuristic: sequential scheduling

We compare the proposed labeling-based approach against a sequential scheduling heuristic. The heuristic is designed to mimic how a driver or dispatcher might plan a fixed itinerary by making decisions incrementally along the route rather than jointly optimizing all charging and HoS decisions. Given a predefined customer sequence , the heuristic constructs a feasible schedule by progressing through the route one leg at a time. For each leg, the heuristic determines whether charging, a mandatory break, or a daily rest is required before continuing. Charging decisions are guided by a battery safety margin and a one-step look-ahead that considers the energy requirements of the subsequent route segment. When charging is required, the heuristic identifies the furthest charging station that can be reached safely on the current leg and charges the vehicle to full capacity. Mandatory breaks and daily rests are inserted whenever required by Hours-of-Service (HoS) regulations, and are synchronized with charging activities whenever both occur at

the same charging station. The heuristic is computationally inexpensive and yields feasible schedules using only local route information, making it a useful benchmark for evaluating the benefits of joint optimization. The detailed scheduling logic is described below.

For each customer-to-customer leg $(A \rightarrow B)$, the scheduler first evaluates whether the current SoC can cover the required energy $E(A! \rightarrow !B)$ while maintaining a safety buffer equal to 30% of battery capacity. A one-step look-ahead is then applied. The vehicle proceeds directly to B if the remaining SoC upon arrival is sufficient either to: (i) complete the next leg $(B \rightarrow C)$ while maintaining the same safety buffer; or (ii) reach a charging station on the leg $(B \rightarrow C)$. Otherwise, the algorithm selects the furthest reachable charging station on $(A \rightarrow B)$, drives there, and recharges the vehicle to full capacity before continuing. If no charging station is reachable on the current leg, the algorithm backtracks to the most recent charging station on the previous leg and revises the schedule accordingly.

HoS constraints are enforced by evaluating each planned route leg against the current break and rest counters. If a mandatory break or daily rest would be due before the leg can be completed, the corresponding interruption (45 minutes or 11 hours, respectively) is inserted. When a stop occurs at a charging station, the scheduler *synchronizes* charging and HoS: it overlaps the required break/rest with the ongoing charge and adds only the excess duration beyond the charging time to the route. The one-step look-ahead used for synchronization also considers the time to the first CS on the next arc, ensuring due stops are anticipated rather than deferred. The full procedure is given in Algorithm 1.

Algorithm 1 Sequential scheduling heuristic

Require: Predetermined route $\mathcal{R} = (c_0, \ldots, c_{N+1})$, leg set $\mathcal{L} = \{(c_i, c_{i+1}) \mid i = 0, \ldots, N\}$

Initialize SoC to full battery capacity, reset all HoS counters, and set $T \leftarrow 0$
for each leg $(c_i, c_{i+1}) \in \mathcal{L}$ **do**
    Check whether the vehicle can reach c_{i+1} while maintaining the safety SoC margin
    if the leg can be completed safely **then**
        Apply one-step look-ahead from c_{i+1}
        if the remaining SoC at c_{i+1} is sufficient to reach c_{i+2} or reach a charging station on leg (c_{i+1}, c_{i+2}) **then**
            Drive to c_{i+1}
        else
            Select the furthest reachable charging station on (c_i, c_{i+1})
            Drive to the selected charging station and charge to full capacity
            Continue to c_{i+1}
        end if
    else
        if a charging station on (c_i, c_{i+1}) is reachable **then**
            Select the furthest reachable charging station on (c_i, c_{i+1})
            Drive to the selected charging station and charge to full capacity
            Continue toward c_{i+1}
        else if backtracking to the most recent charging station yields a feasible continuation **then**
            Revise the previous leg by charging at the most recent charging station
            Retry leg (c_i, c_{i+1})
        else
            return INFEASIBLE
        end if
    end if
    Insert any mandatory break or daily rest required before completing the planned driving
    If the interruption occurs at a charging station, overlap charging and HoS interruption whenever possible
    Apply service at c_{i+1} if c_{i+1} is a customer
    Update SoC, HoS counters, and total duration T
end for
return T

Algorithm 2 Sequential myopic scheduling heuristic

Initialize SoC $\leftarrow B$, HoS counters, $T \leftarrow 0$
for each arc $(A \rightarrow B)$ in $\mathcal{R}$ **do**
    $E_{AB} \leftarrow \alpha \cdot d(A,B)$; $b \leftarrow \beta B$
    Onward look-ahead: compute required ρ_B on $B \rightarrow C$ (energy to next leg or to furthest CS on $B \rightarrow C$, plus b)
    if SoC $\geq E_{AB} + b$ **and** SoC $- E_{AB} \geq \rho_B$ **then**
        Drive(A,B); apply service at B if applicable
    else if SoC $\geq E_{AB} + b$ **and** a CS on $B \rightarrow C$ is reachable after B **then**
    else if a CS on $A \rightarrow B$ is reachable with current SoC **then**
        $s^* \leftarrow$ furthest reachable CS on $A \rightarrow B$
        Drive(A, s^*); ChargeToFull
        DriveWithHoSSync(s^*, B); apply service at B
    else if a CS exists on the previous arc **then**
        backtrack, recharge at last CS, return to A, then retry leg
    else
        return Infeasible
    end if
    Update HoS counters after each drive and service activity
end for
return T
procedure Drive(u, v)
    if HoS limits violated before reaching v **then**
        insert t_{break} or t_{rest}; reset counters
    end if
    $T \leftarrow T + d(u,v)/v_{\text{speed}}$; SoC $\leftarrow$ SoC $- \alpha\, d(u,v)$
end procedure
procedure DriveWithHoSSync(u, v)
    if break/rest due within drive to v (incl. look-ahead to first CS on next arc) **then**
        overlap stop with charging; $T \leftarrow T + \max\{0,\ t_{\text{stop}} - t_{\text{charge}}\}$
    end if
    execute drive $u \rightarrow v$ and update counters
end procedure

6. Computational Results

This section reports the computational results. We first compare partial recharge, full recharge, the greedy heuristic, and the diesel proxy in the base-case setting. We then examine how these differences vary across instance classes and relate them to charging synchronization. Finally, we analyze the sensitivity of the results to battery characteristics, charging infrastructure, and energy consumption.

6.1. *Performance comparison of charging strategies*

Table 4 summarizes the aggregate benchmark comparison for the base-case scenario. The results indicate that the dominant performance effect comes from optimization itself: schedules produced by the exact methods substantially outperform the greedy benchmark, highlighting the benefit of jointly optimizing charging decisions and driver scheduling under the Hours-of-Service regulation. Within this optimized setting, partial recharge provides a further systematic improvement over full recharge, indicating that additional value comes from adjusting charging amounts more precisely to downstream scheduling opportunities.

Table 4: Base-case comparison: partial recharge vs. alternative solution approaches.

Alternative	n	Mean route time (h)		Partial advantage (benchmark − partial)			
		Partial	Alternative	Mean (%)	Mean (min)	Min (min)	Max (min)
Full recharge	43	81.93	82.43	0.61	29.95	4.80	89.39
Greedy	42	83.14	119.15	28.83	2160.35	56.07	4225.01
Diesel proxy	43	81.93	81.30	−0.71	−37.55	−180.00	0.00

Each row compares partial recharge with one alternative on the same test instances (n). Partial advantage: positive values mean partial recharge is faster. Min/Max (min) are the smallest and largest instance-level time differences in minutes.

At the same time, the small average gap to the diesel reference case suggests that, when charging is coordinated effectively with mandatory breaks and daily rests, battery-electric operation can come close to diesel-like performance in the base-case setting. This average result is not uniform across instances, however. Figure 7 shows that the advantage of partial recharge over full recharge varies across the test set, and that the remaining EV–diesel gap is concentrated in a limited number of more difficult cases. The next subsection therefore examines where these gains are largest and how they relate to route structure.

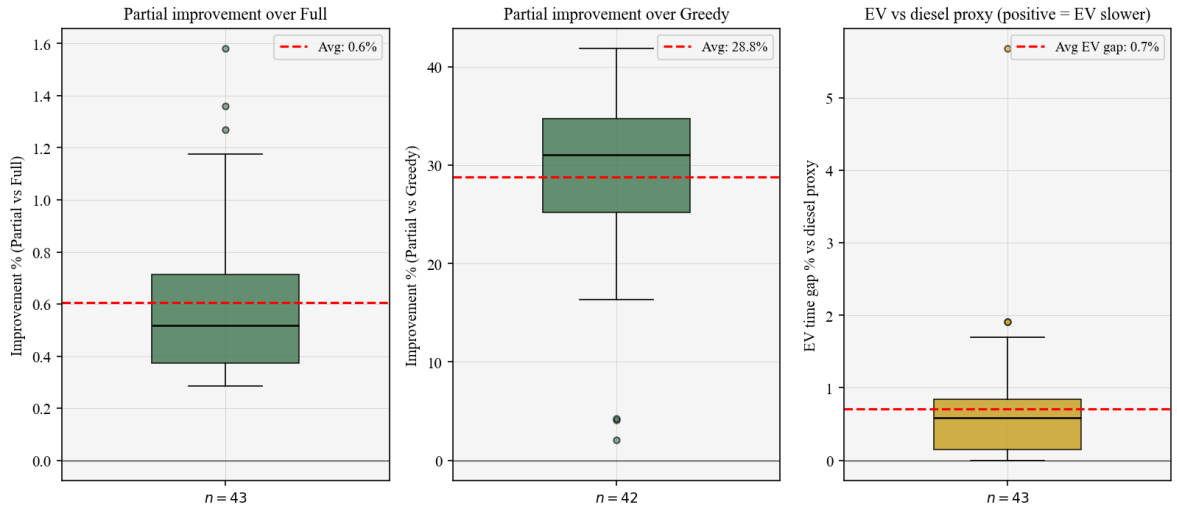


Figure 7: Per-instance comparison of partial recharge against full recharge, the greedy heuristic, and the diesel proxy in the base-case setting.

The benefit of partial recharging increases in absolute terms with route length, although the percentage improvement over full recharging remains modest across all route classes. For short routes, partial recharging reduces the mean route time from 34.32 h to 34.12 h, corresponding to an average saving of 12 min. The mean saving increases to 27 min for medium routes and to around 50 min for long routes. Thus, the additional flexibility of partial recharging becomes more valuable as route duration and the number of charging decisions increase.

This pattern is also visible in the instances with the largest full-versus-partial gaps. The five strongest improvements all occur on long routes, namely L_cl3_cu4_U, L_cl3_cu3_B, L_cl2_cu2_B, L_cl2_cu2_U, and L_cl2_cu3_B, with savings ranging from 62.40 min to 89.39 min. These cases are precisely those in which longer routes create many charging decisions and therefore more scope to improve the schedule by reallocating charging and aligning it with mandatory interruptions. For example, in L_cl3_cu4_U, partial recharge reduces route time by 89.39 min, raises the synchronization share from 58.33% to 78.57%, reduces the number of unsynchronized charging events from 5 to 3, and cuts idle charging time from 122.76 min to 45.98 min, despite increasing the number of charging stops from 12 to 14. By contrast, the smallest gains occur on short routes S_cl3_cu4_U, S_cl2_cu4_U, S_cl2_cu3_B, S_cl3_cu5_B, and S_cl3_cu4_B where the savings are only 4.80 min to 9.60 min. In S_cl3_cu4_U, both strategies already achieve full synchronization, with zero unsynchronized charging and zero idle charging time, so the scope for further improvement is minimal. Detailed per-instance route times and charging-activity statistics are reported in Table C.11.

These results show that the operational value of partial recharge increases with route length and appears to be closely linked to charging synchronization. The next subsection examines this mechanism more directly by relating the observed time savings to the alignment of charging with breaks and daily rests.

6.2. The role of charging synchronization

The route-class results suggest that the value of partial recharge comes not from charging less, but from charging more effectively. Table 5 makes this mechanism explicit. On average, partial recharge uses more charging stops than full recharge (7.53 versus 5.30), yet still achieves a lower mean route time. The improvement arises because these additional charging stops are better aligned with mandatory driver interruptions. In particular, the synchronization share increases from 74.56% under full recharge to 81.89% under partial recharge, while the number of unsynchronized charging events falls slightly from 1.58 to 1.42 despite the larger number of total charges. The main difference comes from break synchronization: partial recharge performs, on average, 2.58 charging stops during breaks, compared with only 0.19 under full recharge, whereas synchronization with daily rests remains essentially unchanged. The same table also clarifies why the greedy benchmark performs poorly. Although the greedy heuristic averages only 4.14 charging stops, none of them are synchronized with breaks or rests, so every charging event adds directly to route duration. The principal benefit of optimization is therefore not simply to reduce the amount of charging, but to embed charging more effectively within time that the driver is required to spend off the road anyway.

Table 5: Charging-synchronization statistics by solution approach in the base-case setting.

Approach	Route time (h)	Charges	Sync. share (%)	Sync. charges	Unsync. charges	Break-sync	Rest-sync
Full recharge	82.43	5.30	74.56	3.72	1.58	0.19	3.53
Partial recharge	81.93	7.53	81.89	6.12	1.42	2.58	3.53
Greedy heuristic	115.90	4.14	0.00	0.00	4.14	0.00	0.00

Entries report averages across the available base-case schedules for each approach. "Sync. charges" and "Unsync. charges" denote charging events that do and do not overlap with mandatory breaks or rests, respectively. "Break-sync" and "Rest-sync" report the average numbers of charging events synchronized with breaks and daily rests.

The same mechanism explains why the gains from partial recharge are largest on long routes. Table 6 compares the five instances with the largest and smallest reductions in route time relative to full recharge. In the largest-gain cases, all of which are long routes, full recharge is associated with comparatively low synchronization and substantial idle charging time. Partial recharge raises the synchronization share from 57.47% to 68.90%, reduces unsynchronized charging from 5.0 to 4.2 events on average, and cuts idle charging time from 94.41 min to 66.60 min, despite using slightly more charging stops. By contrast, the smallest-gain cases are all short routes. There, both strategies already achieve near-complete synchronization, virtually no unsynchronized charging, and negligible idle charging time, leaving little scope for further improvement. Thus, the advantage of partial recharge comes not from fewer charging stops, but from greater flexibility in allocating charging across stops so as to reduce idle and unsynchronized charging when routes become long enough for these effects to matter.

Table 6: Comparison of instances with the largest and smallest partial-recharge gains over full recharge.

Group	Class	Saving (min)	Sync. share (F/P)	Unsync. (F/P)	Idle (F/P)	CS stops (F/P)
Largest gains	Long	74.49	57.47/68.90	5.0/4.2	94.41/66.60	11.6/13.6
Smallest gains	Short	6.72	90.00/93.33	0.2/0.2	3.07/3.07	1.4/2.6

Entries report averages over the five instances with the largest and smallest values of *full recharge route time minus partial recharge route time*. Positive savings therefore indicate that partial recharge is faster. In columns marked "F/P", values are reported as *Full / Partial*.

The cases closest to the diesel benchmark reinforce this interpretation. In the six base-case instances for which the optimized electric schedule matches the diesel benchmark exactly in total route time, every charging stop is synchronized with a mandatory break or rest, and both unsynchronized charging and idle charging are zero. Near-diesel performance is therefore achieved not by avoiding charging altogether, but by absorbing charging fully within the Hours-of-Service structure. Detailed per-instance evidence for these near-diesel cases is reported in Appendix C Table C.10.

6.3. Effects of battery and charging infrastructure characteristics

We next examine how technology and charging-infrastructure assumptions affect the ability of optimized electric-truck schedules to approach the diesel proxy. In this analysis, the benchmark is the full-recharge

electric schedule, and the performance measure is the paired route-time gap relative to the diesel proxy:

$$g_{i,\theta} = 100 * \frac{T_{i,\theta}^{\text{full}} - T_i^{\text{diesel}}}{T_i^{\text{diesel}}},$$

where $T_{i,\theta}^{\text{full}}$ denotes the optimized full-recharge route time for instance i under parameter setting θ, and T_i^{diesel} denotes the corresponding diesel-proxy route time for the same route. Reported averages are computed over EV routes that remain feasible, while feasibility rates are reported separately. Figure 8 shows that EV-diesel gap is increasing when energy demand is high, battery capacity is low, or charging is slow. As the energy-consumption rate increases from 1.12 kWh/km to 2.08 kWh/km, the mean route-time penalty rises from 0.49 % to 3.05 %, reflecting the additional charging burden and reduced flexibility for absorbing recharging within mandatory breaks and daily rests. The opposite pattern is observed for battery capacity and charging power. Increasing battery capacity from 750 kWh to 1000 kWh reduces the mean penalty from 2.52 % to 1.04 %, while increasing charging power from 350 kW to 1000 kW reduces it from 2.73 % to 0.97 %. These reductions indicate that larger batteries and faster charging make it easier to complete required charging within off-duty periods.

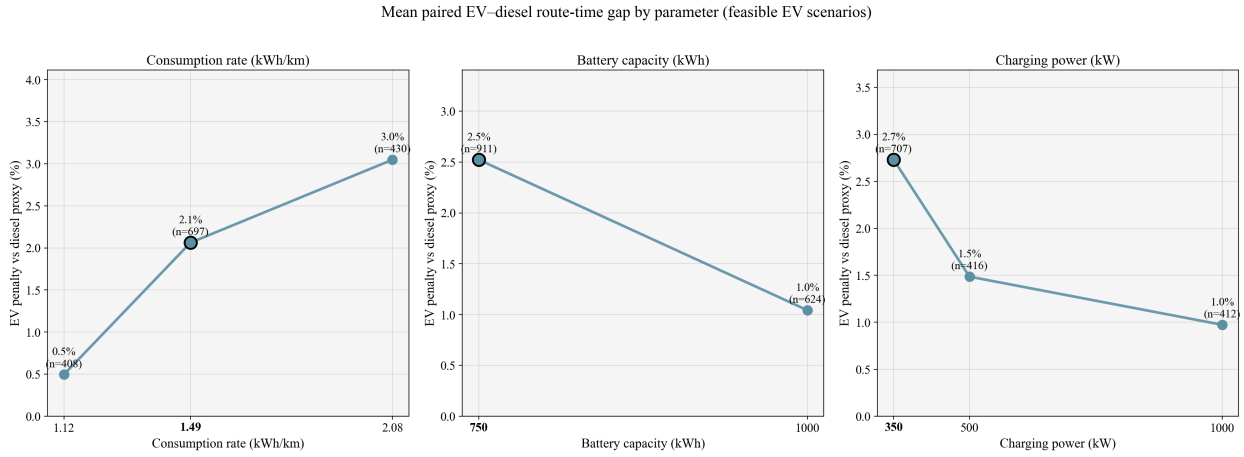


Figure 8: Mean paired route-time penalty of optimized full-recharge electric schedules relative to the diesel proxy, by technology parameter. The highlighted marker indicates the base-case parameter value used in the benchmark comparison.

The charging-pattern indicators in Table 7 support this interpretation. Higher consumption rates and smaller batteries increase the number of charging stops and the number of synchronized charging events. For example, the average number of charging stops rises from 1.44 at 1.12 kWh/km to 3.70 at 2.08 kWh/km. Similarly, the 750 kWh battery case requires 3.60 charging stops on average, compared with 1.94 for the 1000 kWh case. Charging power affects the mechanism differently: higher power reduces the time cost of each charging event, so even when charging remains necessary, the route-time penalty relative to diesel declines substantially.

Figure 9 isolates the role of charging-station spacing. The results show that infrastructure density affects route-time competitiveness before it affects feasibility. When the spacing increases from 60 km to 140 km,

Table 7: Sensitivity summary for optimized full-recharge schedules relative to the diesel proxy.

Parameter	Value	Gap (%)	Gap (min)	Feas. (%)	Charges	Sync.	Unsync.	Break	Rest
Consumption rate	1.12	0.49	13.17	100.0	1.44	1.32	0.13	0.24	1.08
(kWh/km)	1.49	2.06	83.85	99.2	3.31	2.95	0.36	1.2	1.75
	2.08	3.05	75.66	100.0	3.7	3.08	0.63	1.49	1.59
Station spacing	60	1.59	33.81	100.0	2.44	1.81	0.62	0.31	1.5
(km)	80	1.97	42.33	100	2.5	1.94	0.56	0.39	1.56
	100	2.31	50.07	100	2.61	2.06	0.56	0.56	1.5
	120	2.38	108.23	100	4.75	3.88	0.88	1.09	2.78
	140	3.46	171.73	100	5.28	4.78	0.5	2.59	2.19
	160	3.16	141.65	100	4.88	4.47	0.41	1.97	2.5
	180	3.46	166.02	100	5.39	5.24	0.15	2.58	2.67
	200	4.04	196.23	100	5.24	5.06	0.18	3.12	1.94
	220	3.9	186.43	100	5.33	5.06	0.27	2.97	2.09
	240	3.64	167.42	100	4.88	4.62	0.25	2.22	2.41
	260	3.72	183.44	97.0	5.55	5.32	0.23	3.03	2.29
	300	4.95	235.76	84.8	4.48	4.37	0.11	2.7	1.67
Battery capacity	750	2.52	88.59	99.4	3.6	3.21	0.39	1.55	1.66
(kWh)	1000	1.04	25.08	100	1.94	1.59	0.35	0.27	1.32
Charging power	350	2.73	100.86	99.2	3.54	3.23	0.3	1.43	1.8
(kW)	500	1.49	36.47	100	2.46	2.13	0.33	0.73	1.4
	1000	0.97	23.97	100	2.34	1.81	0.53	0.64	1.17

Gap (% and min) is the mean paired route-time penalty of full-recharge vs. diesel-proxy schedules. Feas. reports full-recharge feasibility. Charging columns give mean stop counts; Sync. and Unsync. indicate stops overlapping or not with mandatory breaks or rests. Station-spacing rows use base-case EV settings (1.49 kWh/km, 750 kWh, 350 kW).

the mean full-recharge penalty relative to the diesel proxy rises from 1.59 % to 3.46 %, while all tested scenarios remain feasible. Between approximately 140 km and 240 km, the penalty remains in the range of roughly 3.2 % to 4.0 %, suggesting that the routes can still be completed but charging is increasingly difficult to align with favorable break and rest opportunities. Beyond this range, feasibility begins to deteriorate: the feasibility rate falls to 97 % at 260 km and to 84.8 % at 300 km (Table 7), where the mean penalty among feasible scenarios reaches 4.95 %. It should be noted that the small non-monotonic changes arise because station locations are regenerated for each spacing value, so larger spacings do not uniformly worsen all routes, and because the largest-spacing averages exclude infeasible cases.

7. Conclusion

This paper studied the joint scheduling of charging, mandatory breaks, and daily rests for battery-electric trucks operating on fixed multi-day long-haul routes. We introduced the electric vehicle truck driver scheduling problem as a multi-day extension of the truck driver scheduling setting in which energy

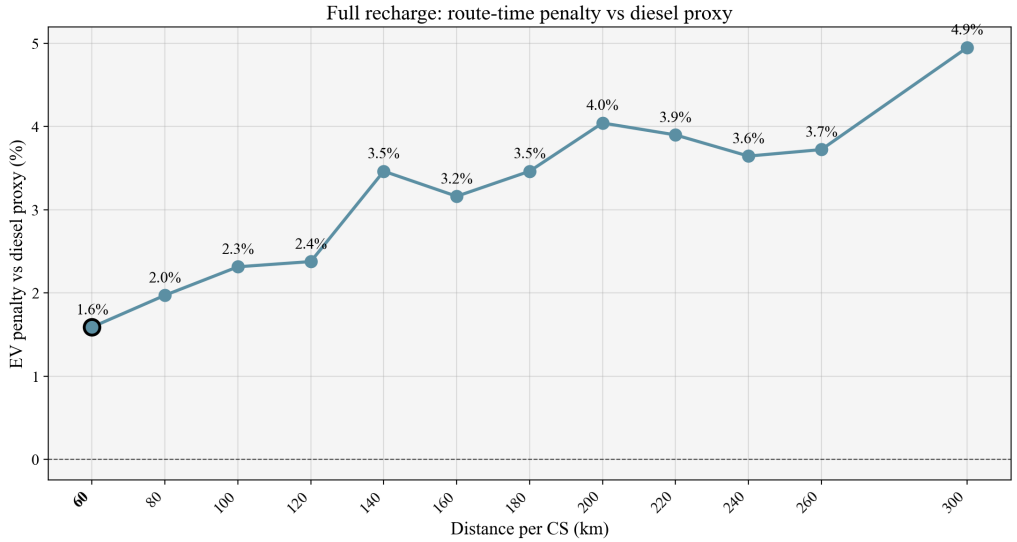


Figure 9: Mean paired route-time penalty of optimized full-recharge electric schedules relative to the diesel proxy as charging-station spacing increases. The highlighted marker indicates the base-case spacing.

replenishment becomes a binding operational resource, and we developed an exact solution approach by extending the parameter-free labeling framework of Goel and Irnich with battery dynamics and charging decisions. The resulting SPPRC formulation captures the interaction between Hours-of-Service compliance and energy feasibility without time discretization, and supports both full- and partial-recharge strategies within a unified framework.

The computational results show that the central operational issue is not charging per se, but charging *in synchrony* with legally mandated interruptions. In the base-case benchmark, partial recharge systematically improves on full recharge, while both exact methods substantially outperform the greedy benchmark. The absolute value of partial recharge grows with route length: although the percentage gain over full recharge remains modest, the time saving rises from about 12 minutes on short routes to about 49 minutes on long routes. Mechanism-based comparisons show that these gains arise because partial recharge allocates charging more flexibly across stops, increases the share of charging synchronized with breaks and rests, and reduces idle and unsynchronized charging on the routes where such effects matter most. Near-diesel performance is achieved precisely in those instances where charging can be fully absorbed into the Hours-of-Service structure.

These findings carry two broader implications. First, the feasibility and competitiveness of long-haul electric trucking depend critically on whether charging opportunities can be embedded within the driver's regulatory timetable rather than added as separate time-consuming stops. Second, the value of partial recharge is highest in technologically tight settings, namely when energy consumption is high, battery capacity is smaller, or charging power is lower. In such regimes, the scheduling problem becomes more constrained, and the ability to tailor charging amounts to downstream breaks and rests becomes materially more valuable. From a planning perspective, this suggests that operational synchronization should be treated

as a first-order design principle alongside battery size and charger deployment.

Several extensions remain relevant for future research. The present study considers fixed route sequences, homogeneous charging technology, deterministic travel and energy consumption. Extending the framework to integrated routing and scheduling decisions, heterogeneous and potentially congested charging infrastructure, nonlinear charging functions, and stochastic travel-time or energy-consumption processes would further strengthen its practical applicability.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI tools in order to suggest alternative formulations for individual sentences. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Full-recharge SPPRC variant and pass-by arcs

needs to be updated validates, after finishing the partial charging description in the main text

This appendix presents the full-recharge variant of the SPPRC framework. The full-recharge strategy is recovered from the partial-recharge model by restricting every charging action to restore the battery to full capacity. Under this restriction, the energy state can be represented by a scalar resource, and additional pass-by arcs are used to preserve feasible continuations when a charging station is traversed without stopping.

Appendix A.1. Network structure with pass-by arcs

Figure A.10 illustrates the extended subnetwork for a route segment from customer n to customer m with a single intermediate charging station s. The corresponding sublegs are (n, s) and (s, m).

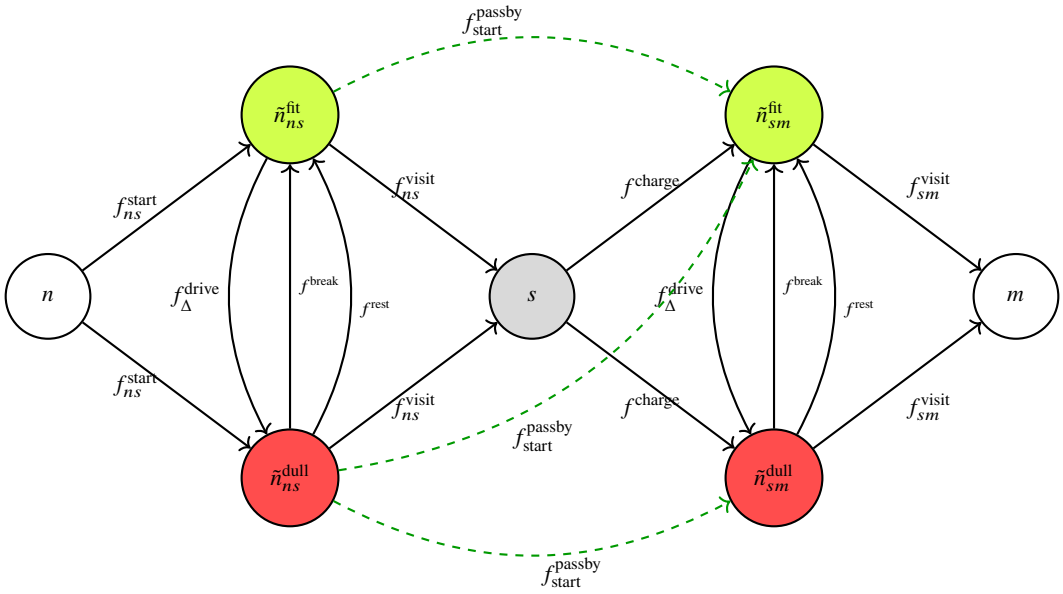


Figure A.10: Extended subnetwork with an intermediate charging station s and pass-by arcs under the full-recharge strategy.

At a charging-station node, the schedule may either initiate a charging activity or continue without charging. Charging is represented through recharge arcs originating from the charging-station node, while pass-by arcs allow the station to be traversed without stopping. Together, these arcs preserve the feasible continuation of both fit and dull driver states while introducing the operational choice of whether and where to recharge.

The pass-by arcs determine how HoS states are propagated when a charging station is traversed without stopping. The fit-to-fit pass-by arc represents a driver who reaches the charging-station location in a state that permits further driving, either because no interruption is due or because the required break or rest has already been completed. If the station is skipped, the driver therefore remains fit and can immediately begin the subleg from the charging-station location to m. A fit-to-dull pass-by arc is unnecessary, since it would require another break or rest before any additional driving; the dull state is reached naturally only

when a driving limit becomes binding on the subsequent subleg. The dull-to-dull pass-by arc preserves an outstanding interruption requirement: after skipping the charging station, the driver remains dull and must take a break or rest before driving toward m. The dull-to-fit pass-by arc is a network connector that preserves the alternative continuations available after the charging-station node, allowing the next subleg to begin with driving and, if necessary, a break or rest to be taken later. It does not reset any HoS resource, so such a continuation is retained only when it is feasible under the current driving and working-time counters.

Appendix A.2. Full-recharge label definition

Under the full-recharge strategy, each charging stop restores the battery to full capacity. The energy state can therefore be represented by a single scalar resource l_{reqEC}, which measures the energy requirement accumulated since the last recharge. Equivalently, the state of charge after any partial path is uniquely determined by $B - l_{\text{reqEC}}$, where B denotes the battery capacity.

A full-recharge label is thus defined as

$$\begin{aligned} l = (&l_{\text{time}},\ l_{\text{timeToNext}},\ l_{\text{drive|R}},\\ &l_{\text{drive|B}},\ l_{\text{elapsed|R}},\ l_{\text{elapsed|B}},\\ &l_{\text{reqEC}}) \end{aligned}$$

The resource l_{reqEC} is bounded by $0 \le l_{\text{reqEC}} \le B$. Its design is inspired by Desaulniers et al. (2016), but we retain an energy-based rather than time-based representation in order to preserve flexibility across different battery capacities, charging powers, and energy-consumption rates.

Appendix A.3. Full-recharge resource extension functions

For the full-recharge strategy, charging at a station cs is represented by an additional REF $f_{\text{cs}}^{\text{charge}}$. This extension resets the accumulated energy requirement l_{reqEC} to zero and increases the elapsed time by the charging duration implied by the current energy deficit and the available charging power. When charging overlaps with a legally admissible break or rest, the corresponding HoS counters are updated consistently so that synchronization between charging and regulatory interruptions is captured directly in the label propagation. In addition, pass-by arcs are modeled through $f_{\text{start}}^{\text{passby}}$, which allows a charging station to be traversed without stopping while preserving all resource values except the remaining travel time to the next physical node. Table A.8 summarizes the resource updates for the REFs together with the full-recharge charging and pass-by extensions.

Appendix A.4. Full-recharge feasibility and dominance

Under the full-recharge strategy, energy feasibility is captured through the accumulated energy requirement since the last recharge. The scalar resource l_{reqEC} must remain within $[0, B]$ and is reset to zero by a recharge extension.

Table A.8: Resource updates for the REFs and the full-recharge charging extension.

Resource	f_{nm}^{start}	$f_{\Delta}^{\text{drive}}$	$f_{\Delta}^{\text{break}}$	f_{Δ}^{rest}	f_{nm}^{visit}	$f_{\text{cs}}^{\text{charge}}$		$f_{\text{start}}^{\text{passby}}$
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_m$	$\begin{cases} l_{\text{time}} & l_{\text{elapsed}\|\text{R}} = 0 \\ l_{\text{time}} + \max\left\{0, \frac{l_{\text{reqEC}}}{CP} - t_{\text{break}}\right\} & l_{\text{elapsed}\|\text{B}} = 0 \\ l_{\text{time}} + \frac{l_{\text{reqEC}}}{CP} & \text{otherwise} \end{cases}$		—
$l'_{\text{timeToNext}}$	t_{nm}	$l_{\text{timeToNext}} - \Delta$	—	—	—	$t_{\text{cs},m}$		$t_{\text{cs},m}$
$l'_{\text{drive}\|\text{R}}$	—	$l_{\text{drive}\|\text{R}} + \Delta$	—	0	—	—		—
$l'_{\text{drive}\|\text{B}}$	—	$l_{\text{drive}\|\text{B}} + \Delta$	0	0	—	—		—
$l'_{\text{elapsed}\|\text{R}}$	—	$l_{\text{elapsed}\|\text{R}} + \Delta$	$l_{\text{elapsed}\|\text{R}} + t_{\text{break}}$	0	$l_{\text{elapsed}\|\text{R}} + s_m$	$\begin{cases} 0 & l_{\text{elapsed}\|\text{R}} = 0 \\ l_{\text{elapsed}\|\text{R}} + \frac{l_{\text{reqEC}}}{CP} & \text{otherwise} \end{cases}$		—
$l'_{\text{elapsed}\|\text{B}}$	—	$l_{\text{elapsed}\|\text{B}} + \Delta$	0	0	$l_{\text{elapsed}\|\text{B}} + s_m$	$\begin{cases} 0 & l_{\text{elapsed}\|\text{B}} = 0 \\ l_{\text{elapsed}\|\text{B}} + \frac{l_{\text{reqEC}}}{CP} & \text{otherwise} \end{cases}$		—
l'_{reqEC}	—	$l_{\text{reqEC}} + \Delta \cdot \alpha$	—	—	—	0		—

Let l_1 and l_2 be two labels ending at the same node. Under the full-recharge strategy, l_1 dominates l_2 if

$$
\begin{aligned}
&l^{(1)}_{\text{time}} \le l^{(2)}_{\text{time}}, \qquad && l^{(1)}_{\text{timeToNext}} \le l^{(2)}_{\text{timeToNext}}, \\
&\mathbf{l}^{(1)}_{\text{drive}} \le \mathbf{l}^{(2)}_{\text{drive}}, \qquad && \mathbf{l}^{(1)}_{\text{elapsed}} \le \mathbf{l}^{(2)}_{\text{elapsed}}, \\
&l^{(1)}_{\text{reqEC}} \le l^{(2)}_{\text{reqEC}}.
\end{aligned}
$$

where

$$
\mathbf{l}_{\text{drive}} = \left(l_{\text{drive}|\text{R}},\, l_{\text{drive}|\text{B}}\right), \qquad \mathbf{l}_{\text{elapsed}} = \left(l_{\text{elapsed}|\text{R}},\, l_{\text{elapsed}|\text{B}}\right).
$$

If these componentwise conditions hold, l_2 can be discarded because it cannot yield a better feasible completion than l_1.

The scalar full-recharge dominance rule is therefore a special case of the partial-recharge dominance rule in the main text, obtained when the time and energy intervals collapse to singletons.

Appendix B. Summary of benchmark instances

We should make this openly available on say Github. Add a link

Table B.9: Key columns only. Distances in km; times in h. Abbreviations: Test = scenario/test name; Class = route class (short/med/long); K = number of clusters; N = total customers; Var = variant (Balanced/Unbalanced); Dist_km = total route distance; Time_h = total route time; Cust/Cl = customers per cluster; AvgCl_km = average cluster distance; AvgIntra_km = average intra-cluster customer distance; AvgInter_km = average inter-cluster distance; AvgSvc_h = average service time per customer.

Test	Class	K	N	Var	Dist_km	Time_h	Cust/Cl	AvgCl_km	AvgIntra_km	AvgInter_km	AvgSvc_h
long_cl1_cu4	long	1	4	-	3929.38	56.1340	[4]	0.00	30.00	0.00	0.2650
long_cl2_cu2_Balanced	long	2	2	Balanced	4485.46	64.0780	[1,1]	674.75	0.00	674.75	0.4579
long_cl2_cu2_Unbalanced	long	2	2	Unbalanced	4472.53	63.8932	[1,1]	972.92	0.00	972.92	0.4207
long_cl2_cu3_Balanced	long	2	3	Balanced	4400.36	62.8623	[1,2]	797.97	37.73	797.97	0.4761
long_cl2_cu4_Balanced	long	2	4	Balanced	4162.88	59.4697	[2,2]	459.89	52.93	459.89	0.2409
long_cl2_cu4_Unbalanced	long	2	4	Unbalanced	4576.22	65.3745	[3,1]	1030.26	103.29	1030.26	0.2304
long_cl2_cu5_Balanced	long	2	5	Balanced	5119.54	73.1362	[3,2]	988.26	73.77	988.26	0.2437
long_cl3_cu12_Balanced	long	3	12	Balanced	5097.76	72.8252	[4,4,4]	810.85	45.65	1081.13	0.1286
long_cl3_cu12_Unbalanced	long	3	12	Unbalanced	4355.65	62.2236	[1,10,1]	647.62	30.00	863.49	0.1340
long_cl3_cu14_Balanced	long	3	14	Balanced	5063.93	72.3419	[5,5,4]	807.74	30.00	1076.99	0.1190
long_cl3_cu3_Balanced	long	3	3	Balanced	5016.56	71.6651	[1,1,1]	818.32	0.00	1091.09	0.3899
long_cl3_cu4_Balanced	long	3	4	Balanced	4155.45	59.3636	[2,1,1]	680.25	62.77	907.00	0.2582
long_cl3_cu4_Unbalanced	long	3	4	Unbalanced	5020.14	71.7163	[1,2,1]	758.57	50.84	1011.43	0.2508
long_cl3_cu5_Balanced	long	3	5	Balanced	5518.38	78.8340	[2,2,1]	894.65	44.15	1192.87	0.2550
med_cl1_cu2	med	1	2	-	2710.05	38.7150	[2]	0.00	84.81	0.00	0.4260
med_cl1_cu4	med	1	4	-	2615.02	37.3575	[4]	0.00	30.00	0.00	0.2420
med_cl2_cu2_Balanced	med	2	2	Balanced	2737.17	39.1024	[1,1]	446.85	0.00	446.85	0.4159
med_cl2_cu2_Unbalanced	med	2	2	Unbalanced	2321.67	33.1667	[1,1]	258.77	0.00	258.77	0.4043
med_cl2_cu3_Balanced	med	2	3	Balanced	3074.20	43.9171	[1,2]	499.00	43.26	499.00	0.4393
med_cl2_cu4_Balanced	med	2	4	Balanced	2413.01	34.4716	[2,2]	436.35	42.90	436.35	0.2751
med_cl2_cu4_Unbalanced	med	2	4	Unbalanced	3105.84	44.3691	[2,2]	681.02	73.82	681.02	0.2348
med_cl2_cu5_Balanced	med	2	5	Balanced	2251.60	32.1658	[3,2]	286.03	47.68	286.03	0.2607
med_cl3_cu11_Balanced	med	3	11	Balanced	2534.30	36.2043	[4,4,3]	332.33	56.27	443.11	0.1281
med_cl3_cu13_Balanced	med	3	13	Balanced	2513.67	35.9096	[5,4,4]	369.80	30.00	493.07	0.1271
med_cl3_cu13_Unbalanced	med	3	13	Unbalanced	3189.23	45.5604	[2,2,9]	437.50	36.03	583.33	0.1135
med_cl3_cu3_Balanced	med	3	3	Balanced	3292.08	47.0297	[1,1,1]	477.29	0.00	636.38	0.4144
med_cl3_cu4_Balanced	med	3	4	Balanced	2265.07	32.3581	[2,1,1]	360.47	142.64	480.63	0.2684
med_cl3_cu4_Unbalanced	med	3	4	Unbalanced	2200.53	31.4361	[2,1,1]	399.06	132.66	532.08	0.2710
med_cl3_cu5_Balanced	med	3	5	Balanced	3472.59	49.6084	[2,2,1]	559.40	34.92	745.87	0.2871
short_cl1_cu2	short	1	2	-	1081.84	15.4549	[2]	0.00	39.79	0.00	0.4019
short_cl2_cu2_Balanced	short	2	2	Balanced	1306.92	18.6703	[1,1]	213.84	0.00	213.84	0.4766
short_cl2_cu2_Unbalanced	short	2	2	Unbalanced	1512.53	21.6076	[1,1]	221.86	0.00	221.86	0.4272
short_cl2_cu3_Balanced	short	2	3	Balanced	1109.01	15.8429	[1,2]	108.55	72.96	108.55	0.3558
short_cl2_cu4_Balanced	short	2	4	Balanced	1221.36	17.4480	[2,2]	162.51	84.56	162.51	0.2685
short_cl2_cu4_Unbalanced	short	2	4	Unbalanced	1408.63	20.1233	[1,3]	159.19	93.35	159.19	0.2364
short_cl2_cu5_Balanced	short	2	5	Balanced	1249.49	17.8499	[3,2]	175.63	35.51	175.63	0.2610
short_cl3_cu10_Balanced	short	3	10	Balanced	1761.42	25.1631	[4,3,3]	260.70	30.00	347.60	0.1333
short_cl3_cu12_Balanced	short	3	12	Balanced	1941.81	27.7401	[4,4,4]	277.09	43.40	369.45	0.1206
short_cl3_cu12_Unbalanced	short	3	12	Unbalanced	1541.88	22.0269	[8,2,2]	215.62	43.30	287.50	0.1268
short_cl3_cu3_Balanced	short	3	3	Balanced	1450.69	20.7242	[1,1,1]	234.78	0.00	313.04	0.4172
short_cl3_cu4_Balanced	short	3	4	Balanced	1270.79	18.1542	[2,1,1]	241.75	106.66	322.33	0.2682
short_cl3_cu4_Unbalanced	short	3	4	Unbalanced	1324.85	18.9264	[2,1,1]	197.50	31.86	263.33	0.2117

short_cl3_cu5_Balanced	short	3	5	Balanced	1876.99	26.8142	[2,2,1]	259.19	68.40	345.58	0.2621

Appendix C. Additional Per-Instance Comparison

Table C.10: Near-diesel diagnostic cases in the base-case setting.

Instance	EV (h)	Diesel (h)	CS	Sync. (%)	Unsync.	Idle (min)
s_cl2_cu4_U	27.86	27.86	3	100.0	0	0
s_cl2_cu3_B	27.19	27.19	2	100.0	0	0
m_cl2_cu2_U	68.95	68.95	6	100.0	0	0
m_cl3_cu4_U	64.10	64.10	5	100.0	0	0
s_cl3_cu10_B	44.20	44.20	4	100.0	0	0
s_cl2_cu4_B	44.19	44.19	2	100.0	0	0

These are the base-case instances for which the optimized electric schedule attains the same total route time as the diesel proxy. In all six cases, every charging stop is synchronized with a mandatory break or rest.

Table C.11: Full- and partial-recharge results from the computational experiments. Times are in hours. Abbreviations: ID = short instance identifier used in the results files; Test = full instance name; F_Time = full-recharge route time; F_CS = number of charging stops under full recharge; F_Sync = number of synchronized charging stops under full recharge; F_Unsync = number of unsynchronized charging stops under full recharge; F_Break = number of full-recharge charging stops synchronized with breaks; F_Rest = number of full-recharge charging stops synchronized with daily rests; P_Time = partial-recharge route time; P_CS = number of charging stops under partial recharge; P_Sync = number of synchronized charging stops under partial recharge; P_Unsync = number of unsynchronized charging stops under partial recharge; P_Break = number of partial-recharge charging stops synchronized with breaks; P_Rest = number of partial-recharge charging stops synchronized with daily rests.

ID	Test	F_Time	F_CS	F_Sync	F_Unsync	F_Break	F_Rest	P_Time	P_CS	P_Sync	P_Unsync	P_Break	P_Rest
S_cl1_cu2_B	short_cl1_cu2	31.73	3	1	2	0	1	31.49	3	1	2	0	1
S_cl2_cu3_B	short_cl2_cu3_Balanced	27.27	1	1	0	0	1	27.19	2	2	0	1	1
S_cl2_cu4_B	short_cl2_cu4_Balanced	44.35	2	2	0	0	2	44.19	2	2	0	0	2
S_cl2_cu4_U	short_cl2_cu4_Unbalanced	27.94	1	1	0	0	1	27.86	3	3	0	2	1
S_cl2_cu5_B	short_cl2_cu5_Balanced	30.88	2	1	1	0	1	30.72	2	1	1	0	1
S_cl3_cu10_B	short_cl3_cu10_Balanced	44.36	2	2	0	0	2	44.20	4	4	0	2	2
S_cl3_cu12_B	short_cl3_cu12_Balanced	43.69	2	2	0	0	2	43.53	4	4	0	2	2
S_cl3_cu12_U	short_cl3_cu12_Unbalanced	30.57	2	1	1	0	1	30.16	2	2	0	1	1
S_cl3_cu3_B	short_cl3_cu3_Balanced	31.35	3	1	2	0	1	30.86	3	2	1	1	1
S_cl3_cu4_B	short_cl3_cu4_Balanced	31.18	2	1	1	0	1	31.02	3	2	1	1	1
S_cl3_cu4_U	short_cl3_cu4_Unbalanced	32.17	2	1	1	0	1	31.85	3	2	1	1	1
S_cl3_cu5_B	short_cl3_cu5_Balanced	31.48	2	1	1	0	1	31.32	3	2	1	1	1
M_cl1_cu2_B	med_cl1_cu2	68.95	6	6	0	0	6	68.95	6	6	0	0	6
M_cl1_cu4_B	med_cl1_cu4	64.26	5	4	1	0	4	64.10	5	5	0	1	4
M_cl2_cu2_B	med_cl2_cu2_Balanced	69.44	4	3	1	0	3	68.95	6	6	0	0	6
M_cl2_cu2_U	med_cl2_cu2_Unbalanced	37.11	2	2	0	0	2	36.62	3	3	0	1	2
M_cl2_cu3_B	med_cl2_cu3_Balanced	47.95	3	3	0	0	3	47.46	4	4	0	1	3
M_cl2_cu4_B	med_cl2_cu4_Balanced	64.09	5	5	0	0	5	63.60	7	7	0	2	5
M_cl2_cu4_U	med_cl2_cu4_Unbalanced	65.70	4	3	1	0	3	65.20	6	5	1	2	3
M_cl2_cu5_B	med_cl2_cu5_Balanced	37.37	2	2	0	0	2	36.88	3	3	0	1	2
M_cl3_cu11_B	med_cl3_cu11_Balanced	47.45	2	2	0	0	2	47.29	4	4	0	2	2
M_cl3_cu13_B	med_cl3_cu13_Balanced	48.39	2	2	0	0	2	48.23	4	4	0	2	2
M_cl3_cu13_U	med_cl3_cu13_Unbalanced	48.98	2	2	0	0	2	48.82	4	4	0	2	2
M_cl3_cu3_B	med_cl3_cu3_Balanced	48.66	4	3	1	0	3	47.84	5	4	1	1	3
M_cl3_cu4_B	med_cl3_cu4_Balanced	38.37	2	2	0	0	2	37.88	3	3	0	1	2

M_cl3_cu4_U	med_cl3_cu4_Unbalanced	33.00	2	2	0	0	2	32.51	3	3	0	1	2
M_cl3_cu5_B	med_cl3_cu5_Balanced	50.26	3	2	1	0	2	49.77	4	3	1	1	2
L_cl1_cu2_B	long_cl1_cu2	74.61	6	4	2	0	4	74.12	7	5	2	1	4
L_cl1_cu4_B	long_cl1_cu4	86.24	5	3	2	0	3	85.75	7	5	2	2	3
L_cl2_cu2_B	long_cl2_cu2_Balanced	75.95	4	3	1	0	3	74.65	7	6	1	2	4
L_cl2_cu2_U	long_cl2_cu2_Unbalanced	76.55	4	3	1	0	3	75.06	6	5	1	2	3
L_cl2_cu3_B	long_cl2_cu3_Balanced	76.02	5	3	2	0	3	74.53	7	5	2	2	3
L_cl2_cu4_B	long_cl2_cu4_Balanced	65.36	4	3	1	0	3	64.05	5	4	1	1	3
L_cl2_cu4_U	long_cl2_cu4_Unbalanced	79.11	6	3	3	0	3	77.81	8	5	3	2	3
L_cl2_cu5_B	long_cl2_cu5_Balanced	88.17	5	3	2	0	3	87.68	7	5	2	2	3
L_cl3_cu12_B	long_cl3_cu12_Balanced	74.79	4	4	0	0	4	74.63	8	8	0	4	4
L_cl3_cu12_U	long_cl3_cu12_Unbalanced	87.07	5	3	2	0	3	86.58	6	4	2	1	3
L_cl3_cu14_B	long_cl3_cu14_Balanced	73.33	4	4	0	0	4	73.16	8	8	0	4	4
L_cl3_cu3_B	long_cl3_cu3_Balanced	88.04	6	4	2	0	4	86.75	7	5	2	1	4
L_cl3_cu4_B	long_cl3_cu4_Balanced	76.94	5	3	2	0	3	75.45	7	5	2	2	3
L_cl3_cu4_U	long_cl3_cu4_Unbalanced	85.90	7	4	3	0	4	84.41	9	6	3	2	4
L_cl3_cu5_B	long_cl3_cu5_Balanced	98.45	6	4	2	0	4	97.96	8	6	2	2	4